

Molecular Detection of Gene Mutation Related to Drug Resistance in *Mycobacterium leprae*

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Abstract

Background: *Mycobacterium leprae*-induced leprosy continues to pose a significant public health threat. Drug-resistant strains pose a major challenge for effective management, necessitating molecular studies to identify resistance-associated mutations and guide appropriate therapy. **Methods:** A cross-sectional analysis of a total of 47 samples, including slit-skin smear and biopsy specimens, were collected along with relevant clinical details. Fifteen samples that tested positive for acid-fast bacilli were further processed. Amplification of *folP1*, *rpoB* and *gyrA* genes was performed using polymerase chain reaction, followed by automated capillary sequencing to identify mutations associated with dapsone, rifampicin, and ofloxacin resistance, respectively, in *M. leprae*. **Results:** Sequencing revealed no *folP1* and *rpoB* gene mutations in any of the 15 isolates sequenced in this study, indicating wild-type status and susceptibility to dapsone and rifampicin, respectively. A mutation was identified at codon 91 (alanine [GCA] → valine [GTA]) in the *gyrA* gene (20%), resulting in an alanine-to-valine change known to cause resistance to ofloxacin. Five samples did not provide adequate chromatogram quality for analysis. **Conclusion:** The study identified mutations in the *gyrA* gene which is associated with ofloxacin resistance in *M. leprae* in Central India. While the absence of resistance to first-line anti-leprosy drugs is reassuring, the emergence of resistance to fluoroquinolones is a cause for concern. Early detection of resistant strains facilitates prompt initiation of drug therapies, reducing their spread and advancing the global leprosy eradication effort.

Keywords: Drug resistance, *gyrA* mutation, *Mycobacterium leprae*, ofloxacin resistance

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INTRODUCTION

Leprosy is a neglected tropical host-pathogen interaction which poses significant public health and clinical challenges.^[1] Although we have already achieved national elimination in 2005, India still accounted for 58% of global new cases in 2019–2020, with 114,451 cases reported.^[2] WHO highlights India as a priority country for contributing to over half of global leprosy cases. Although prevalence has decreased since the 1980s, ongoing transmission necessitates emphasizing early diagnosis, effective treatment, and disability prevention by implementing WHO's "Global Leprosy Strategy 2016–2020" and "Toward Zero Leprosy 2021–2030."^[3]

Leprosy is categorized into five clinical types based on clinical manifestations,^[4] whereas WHO classified it into multibacillary (MB) or paucibacillary (PB) forms based on

skin lesions and bacterial density, in which PB variants are frequently misdiagnosed because they lack distinct and obvious clinical signs.^[5,6] Diagnosis remains complex, particularly in the early stages of the disease, as it requires clinical, histopathological, and specialized laboratory techniques such as bacilloscopy. The diagnosis is based on the disease's distinctive signs and symptoms, which are corroborated by laboratory

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testing.^[7] Leprosy is characterized as either PB (≤ 5 skin lesions) or MB (>5 skin lesions) based on clinical indicators. At the moment, multidrug therapy (MDT) and early case discovery are the primary methods of controlling leprosy and can also be challenging at times due to its variability in presentation and potential to resemble other diseases.^[8,9] Drug resistance forms in *Mycobacterium leprae* have emerged, and failures jeopardize current leprosy control efforts, contributing to a challenging task for its diagnosis.^[10] Mutations in genes such as *folP1* (dapsone resistance), *rpoB* (rifampicin resistance), and *gyrA* (ofloxacin resistance) contribute to resistance, complicating treatment.

An MDT with dapsone, rifampicin, and clofazimine is the standard treatment for leprosy. A common genetic marker for drug resistance in leprosy is the existence of point mutations in genes encoding established drug targets, such as dihydropteroate synthase, DNA gyrase, and the *rpoB* RNA polymerase β subunit.^[11] Studies have shown that resistance to dapsone, rifampicin, and ofloxacin can be due to mutations in the drug-resistance-determining region in the *folP1*, *rpoB*, and *gyrA* genes.^[12] There have been several reports of drug resistance to both dapsone and rifampicin, as well as multidrug resistance, from a variety of nations, including China, India, Brazil, the United States, and Korea.^[10,12-22]

The *folP* gene in *M. leprae* confers dapsone resistance. It has two genetic homologs, *folP1* and *folP2*, with significant sequence homology.^[23] By binding to the dihydropteroate's active site and competing with para-aminobenzoate, dapsone prevents the production of dihydrofolic acid.^[24] The *rpoB* gene, which is involved in rifampicin resistance, inhibits the enzymatic activity of the β -subunit of RNAP holoenzyme by steric occlusion of the 5'-ribonucleotide of the growing RNA chain. This process attenuates bacterial transcription, particularly that of *Mycobacteriaceae*.^[25] A change in the amino acids that border the rifampin-binding pocket of the *rpoB* gene has been shown to cause structural alterations and phenotypic resistance to rifampin in *M. leprae*.^[26] To prevent the development of multidrug-resistant *M. leprae* strains, current leprosy control strategies are based on early detection of cases and treatment with MDT, as recommended by the WHO.^[12] Drug resistance screening is now limited to instances that are either leprosy relapses or do not respond clinically to MDT. The existence or lack of mutations is the basis for the diagnosis, which is supported by clinical data.^[27] Previous studies investigating the structural consequences of point mutations in the *rpoB* gene of *M. leprae* relied on models constructed from distantly related homologs.^[28]

This study uses molecular methods to identify drug resistance-associated gene alterations in *M. leprae* in Central India. We analyzed the DNA sequence of *folP1*, *rpoB*, and *gyrA* genes in *M. leprae* isolates from a tertiary care hospital in Central India and suggest the emergence of drug resistance of *M. leprae*.

The aim of the current study is to fill the gap in prevalence, regional drug resistance mechanism, and clinical importance of *folP1*, *rpoB*, and *gyrA* mutations. With this, we can establish regional mutation patterns, evaluate their effect on antibiotic resistance, and advance molecular diagnostics for precise and

timely detection. We have examined 47 leprosy patients – both new and follow-up cases during the course. DNA was recovered from biopsy samples and slit-skin smears (SSSs) from patients in Central India to detect mutations in *folP1*, *rpoB*, and *gyrA*. Drug resistance gene amplification was done using polymerase chain reaction (PCR) and compared with clinical observations. This study determines the prevalence of drug resistance in *M. leprae*, identifies the drugs to which it is most resistant, and describes the clinical characteristics of patients exhibiting such resistance. This approach facilitates the early identification of drug-resistant cases, enabling timely administration of second-line therapies and helping to prevent the progression of disabilities.

METHODS

Ethical statement

This study received approval from the Institutional Human Ethics Committee-Post Graduate Research of the All-India Institute of Medical Sciences, Bhopal (Ref Number: AIIMS/BPL/IECSR/JAN/23/STS/05, dated August 24, 2023).

Study subjects

This cross-sectional analysis was conducted to detect gene mutations associated with drug resistance in *M. leprae* at a tertiary care hospital and laboratory in Central India. This study was conducted from November to December of 2023, during which 47 participants were enrolled and 15 were selected. All positive SSS/biopsy (if available) cases (new and relapse) along with bacteriological index (BI) ranging from +1 to +6, where acid-fast bacilli (AFB) was seen under ZN Microscopy, ranging between 2 and 90 years of age and belonging to any sex group were included. Inadequate quantity of sample, patients not willing to participate in the study, and no AFB seen in the SSS examination were excluded from the study.

Type of sampling and reasons for selection

The present study used purposive sampling, only confirmed *M. leprae* cases with positive AFB and a defined BI range (+1 to +6) to ensure relevant data for gene mutation analysis were selected. This method was chosen for efficiency and feasibility, given the short study duration (November 2023–December 2023) and the availability of cases at a tertiary care hospital. By focusing on new and relapse cases while excluding AFB-negative and inadequate samples, the study ensures a targeted and high-quality dataset for detecting drug resistance mutations. Out of 47 enrolled participants, only 15 showed AFB positivity, and these 15 samples were further processed for detecting gene mutation related to drug resistance.

Inclusion criteria

1. Patients diagnosed with *M. leprae* infection
2. Cases confirmed through positive SSS and/or biopsy (if available)
3. Both new and relapse cases
4. BI ranging from +1 to +6, where AFB were detected under Ziehl–Neelsen (ZN) microscopy
5. Participants aged 2–90 years and individuals of any sex.

Exclusion criteria

1. Inadequate sample quantity for analysis
2. Patients unwilling to participate in the study
3. No AFB was detected in the SSS examination.

Collection of clinical data

All the patients presenting to outpatient department who fit the inclusion criteria were enrolled in the study after obtaining informed consent. A detailed clinical history was collected, and an examination was done.

Slit-skin smear microscopy

The heat-fixed slides were stained using a modified ZN staining method, and slides were observed under a microscope using immersion oil at $\times 100$.^[29]

DNA extraction

Stained slides were immersed in xylene to remove immersion oil and then treated with 70% alcohol for decolorization. The slides were left to dry at room temperature. Subsequently, 70% alcohol with the chelating agent was poured over the unstained and decolorized slides, and they were gently scraped with the help of a needle.^[30-32] The scrapped material was collected in a microcentrifuge tube. DNA extraction was done using a Qiagen Mini Amp kit (Qiagen, Germany) per the manufacturer's protocol.^[33,34]

Polymerase chain reactions

PCR was performed using specific primer sets [Table 1]. The PCR reaction of 20 μ l was prepared using 2 X Emerald AMP Max hot star PCR master mix (Takara Bio Inc., Japan) [Table 2]. We initially followed the standardized method to perform PCR.^[35] However, we later optimized the reaction conditions, as the optimized protocol yielded better results. The reaction mixture was subjected to initial denaturation at 94°C for 2 min and 45 amplification cycle (94°C for 45 s, 60°C for 1 min and 72°C for 1 min), followed by the final extension step for 8 min at 72°C in thermocycler system (Applied Biosystems) [Table 3].

The amplified products were resolved by agarose gel electrophoresis (1.5% agarose gel) and visualized under an ultraviolet (UV) illuminator using ethidium bromide. Table 4 represents the expected band sizes for different genes.

DNA sequencing

Automated Capillary DNA sequencing was performed using Sanger's method with 3730 DNA Analyser (Applied Biosystems™) 48 capillary sequencer. The raw data obtained were aligned with the wild-type reference sequences of the *M. leprae* TN strain using Codon Code Aligner software to check the presence of any mutations associated with drug resistance.^[11]

RESULTS

In the recent work, a cross-sectional study was conducted in a tertiary care hospital over 2 months in 2023 to detect the gene mutation associated with drug resistance in *M. leprae*. A total of 47 samples were enrolled in the study, out of which 15 samples

Table 1: Primer details

Primer	Sequence	Expected band size (bp)
REP7	TGAGGCTTCGTGTGCTTTGC	450
REP8	ATCTGCGCTAGAAGGTTGCC	450
WHOfolp1 - forward	GCAGGTTATTGGGGTTTTGA	312
folP254 - reverse	CCACCAGACACATCGTTGAC	312
rpoB47	TGGTCCGGGAGCGGATGAC	350
rpoB48	AACCGATCAGACCTATGTTTCG	350
gyrA254 - forward	ATGGTCTCAAACCGGTACATC	254
gyrA254 - reverse	TACCCGGCGAACCAGAAATTG	254

Table 2: Polymerase chain reaction system for primers

Components of reaction	For one reaction (μ L)	Remark
Promega 2X master mix 9 (Takara)	10	Master mix to be used 1X
Forward primer (10 pm)	1.5	Each primer 0.1875
Reverse primer (10 pm)	1.5	Each primer 0.1875 μ L
Nuclease free water	7	For final volume makeup
Template DNA	3	Added in separate area
Total volume	20	For one reaction

Table 3: Steps for polymerase chain reaction

Steps	Temperature (°C)	Time	Cycle
Initial denaturation	95	2 min	1 cycle
Denaturation	94	45 s	45 cycles
Primer annealing	60	1 min	
Initial extension	72	1 min	
Final extension	72	8 min	1 cycle

Table 4: Agarose gel electrophoresis band size

PCR amplicon	Expected band size on gel (bp)
RLEP	450
folP1	312
rpoB	350
gyrA	254

PCR: Polymerase chain reaction

showed positive AFB results and were selected for the gene mutation study. The rest of 32 samples were excluded from the study because no AFB was seen in SSS. The study focused on patients with positive slit-skin smears or biopsy results, spanning ages 2–90 and of any sex, who exhibited a BI ranging from + 1 to + 6 under ZN microscopy. Demographic details of the patients enrolled ($n = 15$) in the study are summarized in Table 5.

A total of 47 patients were initially enrolled in the study, with 15 selected after excluding 32 patients who tested negative for AFB. Among the included patients, 3 were females (20%), and 12 were males (80%). The mean age was 40.8 years. Most patients were between 20–40 and 40–60 years old, comprising

Table 5: Population demographics associated with *Mycobacterium leprae* cases

Demographic parameter	Percentage of patients included (n=15), n (%)
Age	
0–20	1 (6.67)
20–40	6 (40)
40–60	6 (40)
60–80	2 (13.33)
Gender	
Male	12 (80)
Female	3 (20)
Educational status	
Illiterate	3 (20)
Primary school	5 (33.33)
Intermediate	1 (6.67)
High school	2 (13.33)
Graduate	3 (20)
Professional	1 (6.67)

40% of the total samples. In addition, the highest educational level achieved by 33.33% (5 out of 15) of the patients was primary school.

The clinical information of the subjects involved in the study is detailed in Table 6. Relapse cases (80%) outnumbered new cases (20%). The most common site of onset was the upper limbs in 60%, followed by the lower limbs, abdomen, and ear. All cases had skin lesions, with the majority (93.33%) having more than 10 lesions. Sensory loss over skin patches was observed in 12/15 (80%) patients, and nerve involvement was noted in 86.66%. Sensory loss occurred in 66.66% of patients, with the most common pattern being bilateral loss of sensation in the lower limbs, with or without upper limb involvement (40%). The ulnar nerve was the most commonly involved peripheral nerve in 93.33% of cases. Disabilities were noted in two patients, one having a grade 1 and the other a grade 2 disability, while the majority (86.66%) had no disability. Ulceration was present in 33.33% of patients. The motor loss occurred in 26.66% of patients, with motor loss in isolated upper limbs being more common (13.33%) than in isolated lower limbs (6.67%). Type 1 lepra reaction was seen in 20% of patients, and type 2 lepra reaction in 60% of patients. No patients defaulted on treatment. In the study group, 20% were on MDT, and 80% had completed MDT. Among the 15 AFB-positive cases with varying BI values, the most predominant BI values were 2+ and 3+, each observed in 4/15 cases (27%). This indicates that more than half of the patients exhibited moderate levels of bacterial load. Comparing the AFB and PCR positivity between BL and LL types of leprosy, one patient (6.67%) had the BL type, while the remaining 14 patients (93.33%) were classified as LL type; the PCR results for *M. leprae* are presented in Figure 1.

DNA sequencing revealed no mutations in any of the 10 *rpoB* sequences, indicating that all samples were of the wild

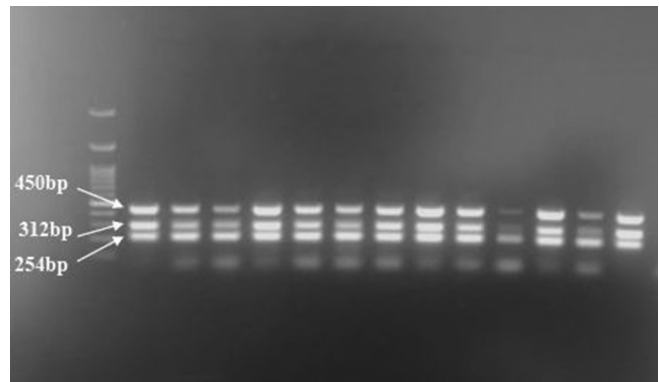


Figure 1: Gel electrophoresis results showing bands corresponding to specific gene amplicons: Lane 1 contained a 100 bp ladder, lanes 2–10 and 12 shows *Rlep* (450 bp), *folP1* (312 bp), *gyrA* (254 bp) bands of amplified genes from the samples, lane 11 and 13 contained diluted sample and lane 14-positive control, and lane 15-negative control

type. Five samples did not yield good-quality chromatograms upon sequencing and, hence, were inadequate for further analysis. Likewise, no mutation was observed among the 10 samples for which *folP1* sequencing results were successfully obtained. However, a *gyrA* gene mutation was identified in 3/15 (20%) samples at the 91 positions, with a substitution from GCA (Ala) to GTA (Val). Mutation at codon 91 is known to be associated with ofloxacin resistance in *M. leprae*.^[11]

As mentioned earlier, *gyrA* mutation was identified in 3/12 of the samples, with one of the three samples included two sample types – tissue and SSS. In contrast, eight samples were found to be wild-type. This distribution of mutant- and wild-type samples in Table 7 highlights the genetic variability in the study population.

DISCUSSION

Leprosy is a persistent infectious condition that can cause permanent harm to the skin, peripheral nerve damage, and possibly disability. It is caused by *M. leprae* or *M. lepromatosis*. Peripheral neuropathy is frequently caused by leprosy, especially in developing or low-income nations. Leprosy impacts the functions of the sensory, motor, and autonomic nerves, resulting in noticeable loss of sensation and weakness of muscles. This is still a public health concern even after decades of successful treatment with MDT.^[36]

Our study aimed to analyze the DNA sequence of a designated region of *M. leprae* involved in drug resistance using *folP1*, *rpoB*, and *gyrA* genes by PCR and to correlate the microbiological findings with clinical manifestations of *leprae*. Our study identified 3/15 (20%) cases with *gyrA* mutation at codon number 91 among leprosy patients in Central India. No *rpoB* gene mutation was found, contrary to our expectations. The absence of resistance to rifampicin is an encouraging finding and suggests that current treatment regimens remain effective in the region. The emergence of

Table 6: Clinical profile of the patients involved in the study (n=15)

Clinical characteristics	n=15, n (%)
Number of new cases	3 (20)
Number of relapse cases	12 (80)
Site of onset	
Upper limb	9 (60)
Lower limb	1 (6.67)
Abdomen	4 (26.66)
Ear	1 (6.67)
Other	0
Number of lesions	
5–10	1 (6.67)
More than 10	14 (93.33)
Loss of sensation over lesion	12 (80)
Nerve involvement	13 (86.66)
Sensory loss	
Total number of patients with sensory loss	10 (66.66)
Only upper limb/limbs	3 (20)
Only left side upper limb (irrespective of lower limb findings)	3 (20)
Only right upper limb (irrespective of lower limb findings)	0
Both upper limbs (irrespective of lower limb findings)	5 (33.33)
Only lower limb/limbs	1 (6.67)
Only left side lower limb (irrespective of upper limb findings)	1 (6.67)
Only right lower limb (irrespective of upper limb findings)	0
Both lower limbs (irrespective of upper limb findings)	6 (40)
Other sites	0
Peripheral nerves	
Ulnar	14 (93.33)
Radial cutaneous nerve	10 (66.66)
Radial	6 (40)
Median	0
Common peroneal	6 (40)
Posterior tibial	3 (20)
Sural	5 (33.33)
Supraorbital	0
Supratrochlear	1 (6.67)
Greater auricular	0
Disability grade	
1	1 (6.67)
2	1 (6.67)
3	0
No disability	13 (86.66)
Presence of ulceration	5 (33.33)
Motor loss	
Present in	4 (26.66)
Only upper limb/limbs	2 (13.33)
Only left side upper limb (irrespective of lower limb findings)	1 (6.67)
Only right upper limb (irrespective of lower limb findings)	1 (6.67)
Both upper limbs (irrespective of lower limb findings)	2 (13.33)
Only lower limb/limbs	1 (6.67)
Only left side lower limb (irrespective of upper limb findings)	0
Only right lower limb (irrespective of upper limb findings)	1 (6.67)
Both lower limbs (irrespective of upper limb findings)	2 (13.33)
Other sites	0

Contd...

Table 6: Contd...

Clinical characteristics	n = 15, n (%)
Defaulters in treatment	0
Patients on MDT	3 (20)
Patients who have completed MDT but not responded	12 (80)
Bacteriological indices	
AFB negative (n=32)	32/47 (68.08)
AFB positive (n=15)	15/47 (31.9)
1+	2/15 (13)
2+	4/15 (27)
3+	4/15 (27)
4+	1/15 (7)
5+	2/15 (13)
6+	2/15 (13)

MDT: Multidrug therapy, AFB: Acid-fast bacilli

Table 7: Analysis of drug resistance genes, associated drugs, and mutation profiles

Drug resistance gene	Drugs associated with the gene	Codon with mutation	Type of mutation	Mutation observed
<i>rpoB</i>	Rifampicin	No mutation observed	Wild type	No mutation identified
<i>folP1</i>	Dapsone	No mutation observed	Wild type	No mutation identified
<i>gyrA</i>	Ofloxacin	Mutation observed at codon 91	GCA [Ala] → GTA [Val]	Mutation identified (point mutation)

GCA: Alanine, GTA: Valine

drug resistance against ofloxacin due to *gyrA* mutation is a cause of concern.

In our study, *gyrA* gene mutation was found in 3/15 (20%) samples at codon number 91 (GCA[Ala] to GTA[Val]). In the study conducted in Korea, 1.92% of cases showed *gyrA* mutation.^[22] In a systemic review, 3.8% of cases out of 2884 from 25 out of 234 cases, 35% resulted in *gyrA* mutation.^[37] In another study conducted in eastern UP, 8/78 samples showed the same mutation in the *gyrA* gene.^[38] Sharma *et al.* concluded in their study that 16/91 *gyrA* gene mutation cases were present in Central India.^[39]

Our study corroborates with the findings in these studies and indicates the emergence of resistance to ofloxacin in India. Our study's cases were associated with paresthesia and bacillary indices of 2+. Due to the small sample size, a significant statistical association could not be obtained.

In a similar study conducted in Korea, mutations in the *folP1* gene of dapsone-resistant *M. leprae* were identified in 20 cases (19.2%). Among these, three cases (2.89%) exhibited mutations in the *folP1* and *rpoB* genes, indicating multidrug resistance to dapsone and rifampin. In comparison, two cases (1.92%) showed mutations in the *folP* and *gyrA* genes,

indicating resistance to dapsone and ofloxacin, respectively. In addition, one case (0.96%) exhibited a double mutation in the *folP1* and the *rpoB* genes.^[22] In various studies conducted in India, *rpoB* gene mutations in leprosy patients were found at 4%(2/50) in a study in Eastern India,^[18] 1.39%^[40] and 3.6% in studies conducted over patients from different parts of India. Our small study sample may have influenced our findings compared to more extensive studies reporting *rpoB* gene mutations in their participants.

In a study conducted in Eastern Uttar Pradesh, no cases (0%) showed *rpoB* mutations. Our findings corroborate the results of this study. The predominant clinical type of leprosy observed in our study was borderline lepromatous (41%), followed by lepromatous leprosy (38.5%), mid-borderline leprosy (15.4%), and histoid leprosy (5.1%). Forty-five cases were new cases, 20 were on MDT, six were defaulter cases, and seven were relapsed cases.^[38] In our study, the majority of cases were classified as lepromatous leprosy (93.33%), with a smaller proportion diagnosed as borderline lepromatous (6.6%). The prevalence of BL and LL types among patients might have influenced our findings, contributing to the similarities observed.

Male *et al.* presented in their study that out of the 214 patients, the clinical presentation according to the Ridley Jopling classification included 62% of the cases to be borderline tuberculoid, 17% were borderline lepromatous, and 7% were lepromatous leprosy cases. Among the 215 participants, 11% (24 cases) exhibited resistance to rifampicin, detected through mutation in the *rpoB* gene through DNA sequencing.^[37] Differences in the clinical picture might contribute to the difference in results. In a separate investigation involving 215 cases, there were 184 relapse cases, 16 new cases, and five defaulters. Among the 16 new cases, mutations were identified in three patients at the previously reported codon positions (424, 425, 437, and 438) and at a novel codon position (411) in the *rpoB* gene. Resistant cases presented with nodules across the body, involvement of multiple nerves, deformities such as claw

hand with finger shortening and bilateral sole, complete anesthesia, and type 1 and type 2 reactions.^[12] In our study, patients had similar symptoms, but none resisted the *rpoB* gene. This suggests that these symptoms may not be linked to drug resistance.

In general, discontinuation of treatments and monotherapy play a significant role in the emergence of drug resistance in leprosy bacillus.^[11] Many factors may have contributed to the lack of detectable drug resistance in our study population, one being the implementation of MDT as the standard treatment protocol for leprosy in India, which may have resulted in the suppression of resistant strains – the MDT recommended by WHO effectively controls the disease, contributing to the global elimination program.^[41] It has shown itself to be an effective technique in the management of leprosy, particularly when patients present early and begin treatment right away.^[42] Furthermore, the absence of detectable drug resistance highlights the importance of continued vigilance and adherence to treatment protocols.

Even though our study provides reassuring evidence of the effectiveness of rifampicin and resistance to ofloxacin in Central India, there are several limitations. The study had a small sample size because of the limited number of available patients over the short duration of 2 months allotted for the study. Reliance on clinical data may have introduced observer bias and limitations in the accuracy of drug resistance estimates. Insufficient amounts of DNA samples from patients in whom only SSSs were available. The low bacillary index might have led to the failure to amplify the *rpoB* gene in 4 patients. In addition, the study's scope was limited to a specific geographic region. Findings may not be generalized to other areas with different epidemiological profiles. Clinical correlation with statistical significance could not be obtained due to the lower sample size. Ofloxacin resistance should be addressed by clinicians in Central India by the use of alternate antibiotics for early diagnosis through systematic susceptibility testing, and antibiotic stewardship to prevent needless prescriptions. It is essential to educate patients on the appropriate use of antibiotics, adjust treatment according to culture results, and regularly monitor treatment results. To effectively manage and mitigate resistance, cooperation with public health authorities on infection control and resistance surveillance is also essential.

Our study did not find any mutations in the *folP1* gene and *rpoB* genes. To characterize drug resistance-associated mutations, subsequent studies should include more relapse cases. There is also a need to carry out the same research over a longer duration for a larger sample size to account for missing the mutation in this study. To improve care and results for impacted patients, it would be beneficial to look into the underlying mechanisms of resistance and assess alternate treatment choices or drug combinations.

CONCLUSION

The study indicates drug resistance to ofloxacin is present in this region, as confirmed by the *gyrA* mutations at the 91 codons, substituting GCA (Ala) to GTA (Val). In our study, we could not detect rifampicin-and dapsone-resistance-associated mutations in the *rpoB* and *folP1* genes, respectively, which is reassuring and affirms the efficacy of rifampicin as a first-line drug in MDT for leprosy, though our findings are based on a small number of new cases of leprosy. The emergence of resistance to second-line anti-leprosy drug ofloxacin is a major concern. It is essential to promote for antibiotic stewardship to prevent the further emergence of resistant strains. Continuous monitoring is necessary to detect any changes in resistance patterns in this region, especially in the cases of ofloxacin resistance reported in Central India, as well as in the case of rifampicin-resistant from other parts of the country.

Outcome of the study

The study did not detect any mutations in *folP1* and in *rpoB* gene sequences, revealing that *M. leprae* isolates remained sensitive to dapsone and rifampicin. A mutation at codon 91 (GCA → GTA) inside the *gyrA* gene appeared in 20% of the analyzed samples, thus conferring ofloxacin resistance. The growing prevalence of fluoroquinolone-resistant infections indicates a developing concern because these antibiotics are mainly prescribed for nontuberculosis bacterial infections. The study demonstrates why molecular surveillance programs are vital for the early detection of resistant strains to help create suitable treatment approaches.

The rationale of the study

The availability of effective treatment for leprosy faces significant challenges because drug resistance continues to develop in public health settings. This study investigates the resistance patterns for dapsone, rifampicin, and ofloxacin by studying alterations in the *folP1*, *rpoB*, and *gyrA* genes of *M. leprae*. The study confirmed no resistance against first-line drugs yet identified a *gyrA* mutation associated with potential ofloxacin resistance. The extensive application of fluoroquinolones demands consistent molecular surveillance to detect resistance in the early stages and stop its spread for global leprosy elimination objectives.

Limitation of the study

The study has a limited sample size (15 cases of *M. leprae* positivity), which limits its wider application. Other alterations linked to resistance may have been overlooked because only the *folP1*, *rpoB*, and *gyrA* genes were examined. Five samples could not be analyzed due to the poor quality of the chromatograms. Insufficient amounts of DNA samples from patients in whom only SSSs were available. The low bacillary index might have led to the failure to amplify the *rpoB* gene in 4 patients. In addition, the study's scope was limited to a specific geographic region. Without long-term monitoring, the cross-sectional design merely provides a glimpse of resistance patterns. In addition, phenotypic drug susceptibility testing was not used to confirm functional alterations. The results

obtained from Central India might not apply to other areas. Future research is required to understand better *M. leprae* drug resistance, geographically diversified samples, and thorough genetic analysis.

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Conflicts of interest

There are no conflicts of interest.

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